Evaluating Structural Encodings in GraphGPS: Zero-Embedding, Ricci Curvature, and Global Graph Statistics

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Abstract

GraphGPS combines local message passing, global attention, and positional or structural encodings (PE/SE) to obtain strong graph learning performance [1]. This final report revisits our mid-term plan and aligns it with the experiments we were actually able to run. We study three families of structural choices on MUTAG, ENZYMES, and NCI1: (i) removing PE entirely through a zero-embedding fallback, (ii) adding discrete Ricci curvature in Forman and Ollivier variants, and (iii) adding coarse global graph statistics. We keep Laplacian PE (LapPE) as the uniform baseline across all comparisons. The results show a mixed picture: zero-embedding hurts ENZYMES and NCI1 but unexpectedly improves MUTAG; Forman curvature is close to the baseline on MUTAG but weaker on the other datasets; Ollivier curvature is unstable and computationally heavier; and global statistics help only modestly, with the clearest gain on NCI1. Together, the findings suggest that structural encodings are helpful, but their value depends strongly on dataset size, topology, and how much information is already present in node features.

1 Introduction

Graph learning models usually need some way to represent structure beyond raw node attributes. In graph Transformers such as GraphGPS, this is typically done by adding positional or structural encodings, which help the model distinguish nodes, preserve geometry, and propagate information over long-range dependencies [1]. Our original mid-term goal was to evaluate a broad set of encodings. For the final report, we narrowed the study to the three experiments that were completed and report them with one consistent baseline: GraphGPS with Laplacian positional encoding (LapPE).

The central question is simple: how much does each encoding really help, and when does it stop helping? We answer this by comparing LapPE against three alternatives. First, we remove PE altogether and use a zero-embedding fallback. Second, we add discrete Ricci curvature, using both the cheaper Forman version and the more expressive Ollivier version. Third, we add graph-level structural statistics such as graph size and density-related summaries. This gives a clean picture of whether a dataset prefers local geometry, coarse global summaries, or does not need much structural input at all.

2 Dataset Structural Profile

Figure 1 summarizes the basic structure of the three datasets.

The EDA suggests three different regimes. MUTAG is the smallest dataset, with only 188 graphs, and the graphs themselves are also small and sparse. ENZYMES has more graphs than MUTAG, but its graphs are the largest on average and also the densest, with the highest average degree. NCI1 is by far the largest dataset in terms of sample count, with 4,110 graphs, but its

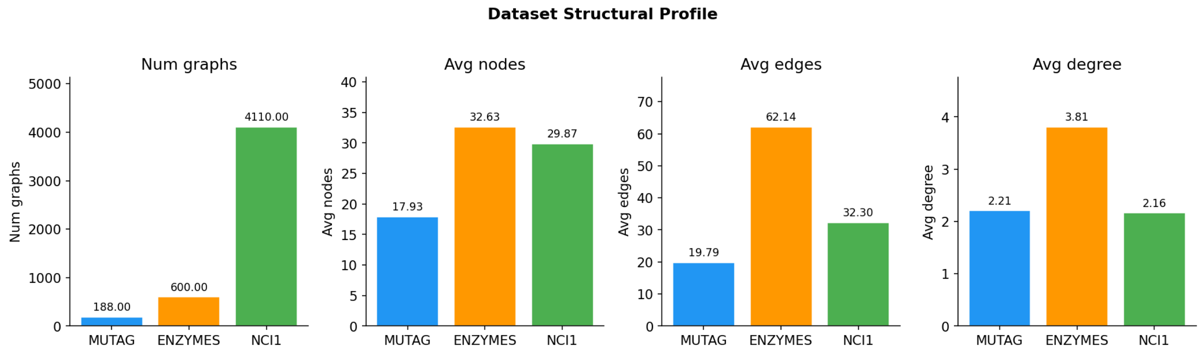

Figure 1: Structural profile of the datasets used in the experiments.

average graph size is moderate compared with ENZYMES. This matters because the usefulness of a structural encoding often depends on whether the task is driven by local motifs, by graph size, or by broader connectivity patterns. In our results, this structural difference shows up clearly: the more complex and data-rich the dataset, the more likely a structural encoding is to help, but only when that encoding matches the topology of the task.

3 Method and Encodings

Let f_θ denote the GraphGPS model. We use LapPE as the common baseline and compare it against three encoding choices.

3.1 Zero-embedding ablation

The zero-embedding setting removes positional encodings and replaces missing structural information with a neutral zero vector. Conceptually, this tests how much the model depends on LapPE rather than on node features and the message-passing backbone alone. If performance drops, then the positional signal is doing real work. If performance stays the same or improves, then LapPE may be unnecessary or may even introduce noise for that dataset.

3.2 Ricci curvature

Ricci curvature is a way of measuring local graph geometry. In graph learning, it is often interpreted as a proxy for bottlenecks, bridges, and communication difficulty. Forman curvature is the simpler and cheaper version; Ollivier curvature is more expressive but significantly more expensive to compute [2, 3]. The motivation is that curvature may expose structures that LapPE alone does not capture, especially in graphs where over-squashing or narrow connections matter.

3.3 Global graph statistics

The third family uses coarse structural statistics such as graph size, edge count, degree pattern, diameter, and related summaries. These features do not describe the full geometry of the graph, but they can provide a useful global prior. They are especially attractive because they are cheap to compute and easy to inject into the model. Their limitation is also obvious: they are too coarse to capture fine local chemistry or connectivity patterns.

4 Results

All deltas below are measured relative to the LapPE baseline. Positive values mean the encoding improved accuracy, while negative values mean it hurt accuracy. We keep the baseline fixed across all tables so that the comparisons stay uniform.

4.1 Table 1: Zero-embedding versus LapPE

Dataset	LapPE baseline	Zero-embedding	Δ (pp)
MUTAG	87.7 ± 2.5	91.2 ± 2.5	+3.5
ENZYMES	68.9 ± 3.4	60.6 ± 3.1	-8.3
NCI1	79.4 ± 1.1	74.2 ± 3.1	-5.2

Table 1: Accuracy comparison between LapPE and zero-embedding. Positive Δ means zero-embedding outperformed LapPE.

The zero-embedding result is the most surprising one in the study. MUTAG improves by 3.5 percentage points when LapPE is removed, which suggests that the positional signal is not always helpful on tiny graphs. One plausible interpretation is that MUTAG is small enough that LapPE adds noise or encourages overfitting rather than providing a useful geometric prior. ENZYMES and NCI1 behave differently: both lose performance without LapPE, with a larger drop on ENZYMES. That means these datasets do benefit from a positional signal, and the backbone alone is not enough. The AUC numbers follow the same pattern: MUTAG improves without LapPE, while ENZYMES and NCI1 improve with LapPE. Overall, the zero-embedding experiment shows that the need for structural encoding is dataset-specific rather than universal.

4.2 Table 2: Ricci curvature versus LapPE

Dataset	Curvature variant	LapPE baseline	Accuracy	Δ (pp)
MUTAG	Forman	87.7 ± 2.5	87.7 ± 2.5	0.0
ENZYMES	Forman	68.9 ± 3.4	64.0 ± 3.8	-4.9
NCI1	Forman	79.4 ± 1.1	78.3 ± 0.4	-1.1
MUTAG	Ollivier	87.7 ± 2.5	84.7 ± 6.0	-3.0
ENZYMES	Ollivier	68.9 ± 3.4	44.2 ± 7.7	-24.7
NCI1	Ollivier	79.4 ± 1.1	80.0 ± 0.8	+0.6

Table 2: Ricci curvature experiments using LapPE as the shared baseline.

Ricci curvature is more mixed than the zero-embedding ablation. Forman curvature stays exactly on par with LapPE for MUTAG, but it trails the baseline on ENZYMES and NCI1. This suggests that the simple curvature signal is not strong enough to beat LapPE in most cases, although it is at least stable on MUTAG. Ollivier curvature is much less consistent: it sharply degrades ENZYMES and slightly improves NCI1. This fits the intuition that Ollivier curvature is richer but also more sensitive and harder to optimize. From a practical standpoint, the result is that curvature is not a plug-in replacement for LapPE; it is a task-dependent feature that sometimes helps, but often does not justify its extra cost.

4.3 Table 3: Global graph statistics versus LapPE

The global-statistics result is the clearest example of a cheap encoding helping only when it matches the dataset. On MUTAG, the gain is negligible, which means these coarse statistics neither help nor hurt much. On NCI1, the improvement is modest but real, suggesting that the graph-level summaries provide useful prior information for a large collection of moderately sized graphs. ENZYMES is the difficult case: performance drops by 16.9 points, which is a strong

Dataset	LapPE baseline	Graph statistics	Δ (pp)
MUTAG	87.7 ± 2.5	87.9 ± 5.8	+0.2
ENZYMES	68.9 ± 3.4	52.0 ± 7.9	-16.9
NCI1	79.4 ± 1.1	80.7 ± 1.5	+1.3

Table 3: Graph-level structural statistics compared against the LapPE baseline.

sign that global summaries are too coarse for this task. In other words, ENZYMES appears to need more detailed structural information than a few aggregate statistics can provide.

5 Discussion

Taken together, the results support three main conclusions.

First, LapPE is a strong default baseline, but it is not always the best choice. The zero-embedding result on MUTAG shows that positional information can be unnecessary or even harmful on small datasets.

Second, curvature features are not universally beneficial. They are theoretically appealing because they encode local bottlenecks and connectivity shape, but in practice their usefulness depends on whether the downstream task actually depends on that kind of geometry. Forman is more stable; Ollivier is more variable.

Third, simple graph statistics are useful only when the dataset has a clear global-structure signal. They help a little on NCI1, do not matter much on MUTAG, and fail badly on ENZYMES. This is an important practical lesson: cheap features are attractive, but they are not automatically good features.

The EDA helps explain these patterns. MUTAG is tiny, so a richer positional encoding may overfit. ENZYMES is larger and denser, which means local and mid-range structure matter more than a few global summaries. NCI1 contains far more graphs, so the model can benefit from even a small structural prior because the dataset is large enough for that signal to be learned reliably.

6 Conclusion

This final report refines the mid-term plan into a smaller but cleaner empirical study. Using LapPE as a shared baseline, we compared zero-embedding, Ricci curvature, and global graph statistics on three benchmark datasets. The experiments show that no single structural encoding is best everywhere. LapPE remains a robust baseline, but zero-embedding can win on small graphs, curvature is useful only in a limited and dataset-dependent way, and global statistics are cheap but often too coarse. The main takeaway is that structural encodings should be matched to the dataset rather than added by default.

References

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Links: Github